Chapter 9

Dentin Hypersensitivity

"The sensitivity of men to small matters, and their indifference to great ones, indicates a strange inversion".

—Blaise Pascal

Dentin hypersensitivity is a common clinical condition that is difficult to treat because the treatment outcome is not consistently successful. Patients may complain of discomfort when teeth are subjected to temperature changes, osmotic gradients such as those caused by sweet or salty foods or even tactile stimuli. The cervical area of teeth is the most common site of hypersensitivity.

Definition

According to Holland et al.,[1] *dentin hypersensitivity* is characterized by short, sharp pain arising from exposed dentin in response to stimuli typically thermal, evaporative, tactile, osmotic or chemical and which cannot be ascribed to any other form of dental defect or pathology.

Etiology

Dentin hypersensitivity is a multifactorial condition and is not fully understood. However, there is a definite change in the structure of dentin in the affected area. These areas contain a larger number of patent dentinal tubules with a larger tubular diameter than the normal unaffected areas. The following three theories have been proposed to explain this condition:

Direct Innervation Theory

This theory[2] postulates that direct mechanical stimulation of exposed nerve endings at the dentinoenamel junction (DEJ) is responsible for dentinal hypersensitivity.

The major shortcoming of this theory is that there is insufficient evidence to prove that the outer dentin that is most prone to be sensitive is well innervated.

Odontoblast Receptor Theory

This theory[3] proposes that the odontoblasts themselves act as neural receptors and relay the signal to the nerve terminal.

The major shortcoming of this theory is that there is no evidence to demonstrate synapses between odontoblasts and nerve terminals.

Hydrodynamic Theory

This is the *most accepted theory* to explain the mechanism behind dentin hypersensitivity.

The *hydrodynamic theory* proposed by *Brannstrom*[4] suggests that dentin hypersensitivity is due to the hydrodynamic fluid movements occurring across exposed dentin with open tubules which in turn mechanically activates the nerves present at the inner ends of the dentin tubules or in the outer layers of the pulp (Figs. 9.1 and 9.2).

This theory postulates that the dentin tubules, which are open and wide, contain fluid. Various stimuli (i.e. thermal, tactile, chemical or osmotic changes) displaces this fluid in the dentinal tubules in either an inwardly or outwardly direction. The movement of this liquid stimulates the odontoblastic processes, and the subsequent mechanical disturbances stimulates baroreceptors (a nerve receptor sensitive to pressure) that leads to neural discharges (depolarization). This neural pulpal activation is perceived as pain by the patient.

Box 9.1 explains the hydrodynamic theory of the mechanism of dentin hypersensitivity.

The causes for dentinal exposure could be the result of one of the following processes[5]:

I. Anatomical characteristics of the cementoenamel junction (CEJ)
II. Loss of the enamel covering the crown of the tooth

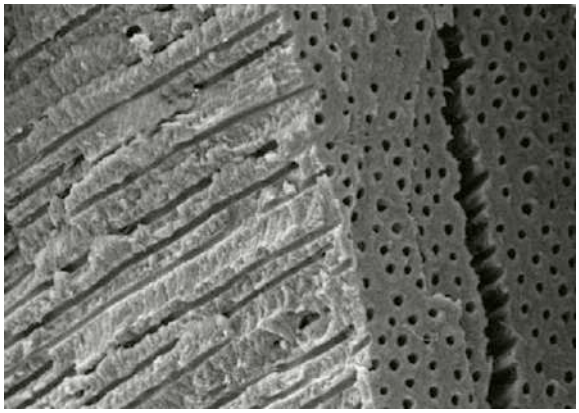


Figure 9.1 Dentinal tubule: cross-section of dentin showing dentinal tubules and their orientation.
(From Suresh Chandra, Gopikrishna V: *Grossman's endodontic practice*, ed 13, 2014, Wolters Kluwer-Lippincott.)

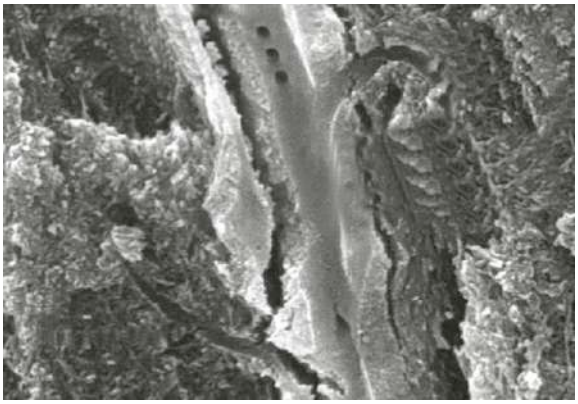


Figure 9.2 Dentinal tubules are normally filled with odontoblastic processes and dentinal fluid. Hydrodynamic fluid movement can result in pain transmission.
(From Suresh Chandra, Gopikrishna V: *Grossman's endodontic practice*, ed 13, 2014, Wolters Kluwer-Lippincott)

Box 9.1 Hydrodynamic theory of dentin hypersensitivity

Stimulus
↓
Fluid flow in exposed open dentinal tubules
↓
Stimulation of intra-tubular baroreceptors
↓
Pain

III. Denudation of the root surface due to loss of cementum and overlying periodontal tissues

Box 9.2 provides a list of aetiological and predisposing factors associated with dentin hypersensitivity.

Box 9.2 Aetiological and predisposing factors associated with dentin hypersensitivity[6]

Loss of enamel
Gingival recession
Attrition (Figs. 9.3 and 9.4)
Erosion (intrinsic and extrinsic) (Fig. 9.5)
Tooth malposition
Denudation of cementum (Fig. 9.6)
Thinning, fenestration, absent buccal alveolar bone plate
Abrasion
Abfraction (Fig. 9.7)
Periodontal disease and its treatment
Periodontal surgery
Patient habits

Clinical Features

I. Pain is the most common clinical feature associated with dentin hypersensitivity.
II. The intensity of pain varies from mild discomfort to severe sensitivity.
III. The pain is typically rapid in onset, sharp in character and is of short duration.[5]
IV. External stimuli which can elicit the expression of this condition include:
   i. Thermal stimuli
      – Hot/cold food and beverages
      – Cold blast of air
   ii. Osmotic stimuli
      – Sweet food
   iii. Acidic stimuli
      – Citrus fruits
      – Acidic beverages
      – Medicines



Figure 9.3 Attrition of mandibular posteriors causing dentinal hypersensitivity.



Figure 9.4 Attrition of mandibular anteriors causing dentinal hypersensitivity.



Figure 9.5 Cervical erosive lesions causing hypersensitive dentin.



Figure 9.6 Denuded cementum is a common cause for dentinal hypersensitivity.

iv. Mechanical stimuli
   - Toothbrush
   - Dental instruments

v. The most commonly involved teeth are
   i. Buccal surfaces of premolars
   ii. Facial surfaces of incisors



Figure 9.7 Abfractive lesions in cervical region causing sensitivity.

Management of Dentin Hypersensitivity

Diagnosis

I. A thorough patient history and clinical examination is mandatory.
II. Diagnosis is confirmed by using a jet of air or by using a diagnostic probe on dentin in a mesiodistal direction in the area where the patient complains of pain.[5]

Clinical Management

Removal of Etiological Factors

The removal of the following aetiological factors can prevent dentin hypersensitivity from recurring[7]:

I. Improper tooth brushing technique.
II. Poor oral hygiene.
III. Premature contacts.
IV. Gingival recession.
V. Endogenous and exogenous acids.

Patient Education

It is important to counsel the patient in order to modify their dietary habits. Efforts should be made to teach them a proper tooth brushing technique and achieve plaque control.

Treatment Strategies

The various clinical methods of management of dentin hypersensitivity are given in Box 9.3.

Box 9.3 Treatment strategies for dentin hypersensitivity

1. Desensitizing the nerve
   i. 5% Potassium nitrate
2. Occluding the dentinal tubules
   i. Recent advances
      – Arginine
      – CSPS: Calcium Sodium Phospho Silicate
   ii. Salts or ions
      – 0.4% Stannous fluoride
      – 2% Sodium fluoride
      – Potassium oxalate
      – Strontium chloride
      – Calcium carbonate
   iii. Protein precipitates
      – CPP–ACP (Casein Phosphopeptide - Amorphous Calcium Phosphate)
      – Silver nitrate
      – Glutaraldehyde
      – Zinc chloride
3. Dentin adhesives
   i. Dentin bonding agents
   ii. Composites
   iii. Glass ionomers
4. Crown placement
5. Periodontal grafting
6. Lasers

Potassium nitrate desensitization

I. Daily use of 5% potassium nitrate-containing toothpaste for 4 weeks is an effective way to desensitize dentin.[8]
II. Potassium nitrate increases the extracellular potassium ion concentration and thus depolarizes the nerve. This prevents the transmission of pain signals to the brain.[9]

Clinical Notes

Potassium nitrate has also been proved to be effective in controlling post-bleach sensitivity by employing it as an in-office 10–30 min therapeutic gel application before and after the bleaching procedure.

Occluding the dentinal tubules

Occluding the distal terminal ends of exposed dentinal tubules is another method of managing dentin hypersensitivity.

I. Fluorides decrease the dentinal permeability by precipitating calcium fluoride crystals inside the dentinal tubules.[7]
II. Fluorides are used in the form of *2% sodium fluoride* or *0.4% stannous fluoride.*
III. *Potassium oxalate* has also been proved to reduce dentin hypersensitivity by forming calcium oxalate crystals inside the dentinal tubules.[7]
IV. Casein is a milk protein, which has been used to develop *casein phosphopeptide* (*CPP*) that gets attached to *amorphous calcium phosphate* (*ACP*).[10] This **CPP–ACP** complex has been proved to enable early enamel remineralization in white spot lesions. They also have shown to have potential to prevent and treat dentin hypersensitivity.
V. **Arginine** is being incorporated in a number of recently developed toothpastes (Colgate Sensitive Pro-Relief, Colgate-Palmolive Company, NY, USA). The combination of arginine with calcium carbonate and phosphate results in precipitate formation on the open dentinal tubules.[11] In addition to toothpastes, it is available in several forms like:
   i. Arginine in-office paste (8%)
   ii. Arginine mouth wash (0.8%)
VI. **Calcium Sodium Phosphosilicate (CSPS)**: *NovaMin* is the trade name for a CSPS bioactive glass. It acts by occlusion of dentinal tubules by precipitating calcium and phosphate. Calcium and phosphate ions react with oral fluids to form hydroxy-carbonate apatite layer that is similar to natural tooth structure.[12] It is available in the form of toothpaste (5% CSPS) and prophylaxis paste (15% CSPS), for example, Sensodyne Repair and Protect.

Dentin adhesives

I. The use of dentin adhesives to treat hypersensitive dentinal surfaces has gained popularity. Reductions in sensitivity can result from formation of resin tags and a hybrid layer when a dentin adhesive is used.[13]
II. The primers of the multi-bottle adhesive system All-Bond 2 have a desensitizing effect, even without consistent resin tag formation.[14]

Clinical Notes

- In a clinical study, using the primer of the original GLUMA adhesive system (an aqueous solution of 5% glutaraldehyde and 35% HEMA, currently marketed as GLUMA Desensitizer [Heraeus Kulzer]), the authors concluded that GLUMA primer reduced dentin sensitivity through a protein denaturation process with concomitant changes in dentin permeability.
- Glutaraldehyde has long been used as a fixative that cross-links proteins.[15,16]

Lasers

I. Recently, lasers have also been shown to be effective in managing dentin hypersensitivity.
II. Nd-YAG lasers have shown to effectively occlude dentinal tubules.[17,18]

Summary

Effective management of dentin hypersensitivity should incorporate a detailed clinical history along with identification of aetiological factors. Clinicians have used many materials and techniques to treat dentin hypersensitivity, including specific dentifrices, laser irradiation, dentin adhesives, antibacterial agents, aldehydes, fluoride rinses, fluoride varnishes, calcium phosphate, potassium nitrate and oxalates among others. A combination of patient education and effective treatment strategy is effective in alleviating the pain and discomfort associated with dentin hypersensitivity.

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